3)

3.1) a - $\{1, -1, 2, 0\}$

b - $\{1, 0, 1/4\}$

c - $\{1, -1, 0\}$

d - $\underbrace{a^2 \in A}_{-1,1,0}$ e $\underbrace{a \geq 0}_{1,2,0,1/4}$ $\underbrace{x = \sqrt{a}}$

$\underbrace{-1,1,0 \qquad 1,2,0,1/4}_{1,0}$

$\{1, 0\}$

e - $\{1, 0, 4\}$

f - $\{\pm 1, \pm 1/2, \pm\sqrt{2}, 0\}$

3.2) a - $A = \{x \in \mathbb{R} \mid |x| = 1\}$

b - $B = \{c \in \mathbb{N} : \forall_{a,b \in \mathbb{N}}\ c = ab \wedge a \neq 1 \Rightarrow b = 1\}$

c - $C = \{x \in \mathbb{R} \mid x = 3m, m \in \mathbb{N}\}$

d - $D = \{b \in \mathbb{N} \mid \exists_{x \in \mathbb{N}} (2 \leq x \leq 5, b = x^2)\}$

3.3) a - $\{x \in \mathbb{R} \mid x^2 - 3x + 2 = 0\}, \{1, 2\}$

b - São todos iguais

c - $\varnothing$, $\{\}$

d - $\{1, -1\}$ e $\{x \in \mathbb{R} \mid x^2 = 1\}$

3.4) a - V, porque 5 é um dos elementos do conjunto A

b - F, porque $\{5\}$ não está contido em A

c - V

d - F

e - V

f - V

g - F

h - F

3.5) a - V

b - F

c - F

d - V

e - V

f - V

g - F

h - F

3.6) a - V

b - V

c - V

d - F

3.7) $A \subseteq B \subseteq C$

a, e, f

3.8) a -

$A = \{1, 2, 3\}$ $B = \{1, 2, 3, 4, 5\}$

b- $A = \{1,2\}$

$B = \{\{1,2\},3,4\}$

c- $A = \{1,2\}$

$B = \{3,4\}$

d- $A = \{1,2\}$

$B = \{1,2,\{1,2\},3,4\}$

3.9) $C = \{4,16,36,64\}$

$A \cup C = \{2,4,6,8,16,36,64\}$

$A \cup B = B$ fois $A \subseteq B$

$C \cup B = B$ fois $A \subseteq B$

$A \cup A = A$

$A \cap B = A$

$B \cap B = B$

$B \cup C \cup A = B$

$C \setminus A = \{2,16,36,64\}$

$A \setminus B = \emptyset$

$B \setminus A = \{x \in \mathbb{N} \mid \exists y > 4 \; x = 2y\}$

3.10) a- $\{0 < x < 6\}$

b- $\{2 < x < 3\}$

c- $\{x \in X : -11 < x \leq 0 \lor 3 \leq x < 11\}$

d- $\{3 \leq x < 6\}$

e - $\{ x \in X : -11 < x \leq 2 \ \vee \ 6 \leq x < 11 \}$

f - $A \cap (B \cup C) = \underbrace{(A \cap B)}_{2 < x < 3} \cup \underbrace{(A \cap C)}_{0 < x < 1}$

$= x \in]0,1[\vee]2,3[$

g - $\underbrace{(A \cap B)}_{2 < x < 3} \cup \underbrace{(A \cup C)}_{-1 < x < 3}$

$x \in]-1,3[$

h - $\underbrace{(A \cap B)}_{2 < x < 3} \cup \underbrace{(A \cap C)}_{0 < x < 1}$

$=]0,1[\cup]2,3[$

3.11) a - $A \cup A = A$

Dado x

$x \in A \cup A \Leftrightarrow x \in A \vee x \in A \Leftrightarrow x \in A$

logo

$A \cup A = A$

c - $A = (A \cap B) \cup (A \setminus B)$

$x \in (A \cap B) \cup A \setminus B \Leftrightarrow x \in (A \cap B) \vee x \in A \setminus B$

$\Leftrightarrow (\underline{x \in A} \wedge x \in B) \vee (\underline{x \in A} \wedge x \notin B)$

$\Leftrightarrow x \in A \wedge \underbrace{(x \in B \vee x \notin B)}_{\text{tautologia}}$

$\Leftrightarrow x \in A$

b- $x \in A \setminus B \Leftrightarrow x \in A \wedge x \notin B \Rightarrow x \in A$

d- $A \cap (B \setminus C) = (A \cap B) \setminus C$

$x \in A \cap (B \setminus C)$

$\Leftrightarrow x \in A \quad \wedge \quad x \in (B \setminus C)$

$\Leftrightarrow x \in A \quad \wedge (x \in B \quad \wedge x \notin C)$

$\Leftrightarrow (x \in A \quad \wedge x \in B) \wedge x \notin C$

$\Leftrightarrow x \in (A \cap B) \quad \wedge x \notin C$

$\Leftrightarrow x \in (A \cap B) \setminus C$

3.12) a- $\forall_{A,B,C} \quad C \subseteq A \cup B \Rightarrow C \subseteq A \wedge C \subseteq B$

$A = \{1\}$, $B = \{1,2\}$, $A \cup B = \{1,2\} = C$

mas $C \not\subseteq A \quad \wedge \quad C \not\subseteq B$

b- $\forall_{A,B,C} \quad C \subseteq A \vee C \subseteq B$

$\Leftrightarrow C \subseteq A \cup B$

Se $C \subseteq A$ como $A \subseteq A \cup B$ então $C \subseteq A \cup B$
Se $C \subseteq B$ como $B \subseteq A \cup B$ então $C \subseteq A \cup B$

e- $\forall_{A,B,C} \quad A \subseteq C \wedge B \subseteq C \Rightarrow A \cup B \subseteq C$

$x \in A \cup B \Leftrightarrow x \in A \vee x \in B$

Se $x \in A$, Como $A \subseteq C$ então $x \in C$

Se $x \in B$, Como $B \subseteq C$ então $x \in C$

d- $\forall_{A,B,C} \quad A \cup B \subseteq C \Rightarrow A \subseteq C \wedge B \subseteq C$

Como $A \subseteq A \cup B$ e $A \cup B \subseteq C$, então $A \subseteq C$

Como $B \subseteq A \cup B$ e $A \cup B \subseteq C$, então $B \subseteq C$

3.20) $\# A = m \qquad \# P(A) = 2^m \qquad \#(A \times A) = m^2$

$\#(P(A) \times P(A)) = 2^m \times 2^m = 2^{2m}$

$\# P(A \times A) = 2^{(m^2)} \neq 2^{2m}$

$\forall_{m \geq 3} \; \# P(A \times A) \geq \# P(A) \times P(A)$